STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-QC-COMPARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4961-4968 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 4970-4994, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_algebraic-stacks-sheaves.json (pb=1217, stmt 4961-4968, proof 4970-4994).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DDH: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4994–4994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Z0: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4994–4994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-restriction: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1185–1200. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZS: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4994–4994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GH: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4994–4994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. The comparison morphism $\epsilon : \mathcal{X}_{affine,étale} \to \mathcal{X}_{affine}$ satisfies the assumptions and conclusions of Cohomology on Sites, Lemma 0GZS.*

Proof. Assumption (1) holds by definition of $\mathcal{X}_{affine}$. For condition (2) we use that for $x \in \mathrm{Ob}(\mathcal{X})$ lying over the affine scheme $U = p(x)$ we have an equivalence $\mathcal{X}_{affine,étale}/x = (Aff/U)_{étale}$ compatible with structure sheaves; see discussion in Section section-restriction. Thus it suffices to show: given an affine scheme $U = \mathrm{Spec}(R)$ and a complex of R-modules $M^\bullet$ the total cohomology of the complex of modules on $(Aff/U)_{étale}$ associated to $M^\bullet$ is quasi-isomorphic to $M^\bullet$. This follows from a combination of: Derived Categories of Schemes, Lemma 06Z0 (total cohomology of complexes of modules over affines in the Zariski topology), Derived Categories of Spaces, Remark 08GH (agreement between total cohomology in small Zariski and étale topologies for quasi-coherent complexes of modules), and Étale Cohomology, Lemma 0DDH (to see that the étale cohomology of a complex of modules on the big étale site of a scheme may be computed after restricting to the small étale site). □